GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for Electrical Equipment**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ELE-SPC-002
Document Revision	R01
Document Status	Issued for Review
Originator	Vincent Ani
Security Classification	Restricted
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Specification for Electrical Equipment

R01	28.07.22	Issued for Review	N. Kari	V. Ani	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	28.07.22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
   for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited		
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A	
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-ELE-SPC-002	

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001	Load & Electrical Equipment List – BBOU Load & Electrical Equipment List - ELEP Load & Electrical Equipment List - AKAB
EA-BFPCL-GPMC-GGPL-FEED-ELE-DSH-002	Data Sheets for Electrical Equipment
EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-001	Finalization of Electrical Power Requirement and Power Supply
DEP. 82.00.10.10-Gen	Project Quality Assurance
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
A	Ampere
AC	Alternating Current
AEP	Amukpe Escravos Pipeline
ATEX	Atmosphere Explosive
DC	Direct Current
EMC	Electromagnetic Compatibility
EOT	Escravos Oil Terminal
FAT	Factory Acceptance Test
FEED	Front End Engineering Design
FOT	Forcados Oil Terminal
FSO	Floating Storage and Offloading
GRP	Glass Reinforced Polyester
Hz	Hertz
IEC	International Electro Technical Commission
IP	Ingress Protection
ISO	International Standard Organization
LV	Low Voltage (phase-to-phase voltage of between 100V and 1,000V)
MCB	Miniature Circuit Breakers
MCCB	Moulded Case Circuit Breaker
OML	Oil Mining Lease
RCD	Residual Current Device
RTU	Remote Terminal Unit
SI	International System
SPN	Single Pole and Neutral
SWA	Steel Wire Armoured
TEP	Trans Escravos Pipeline
TFP	Trans Forcados Pipeline

VAC	Voltage (Alternating Current)
VT	Voltage Transformer

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

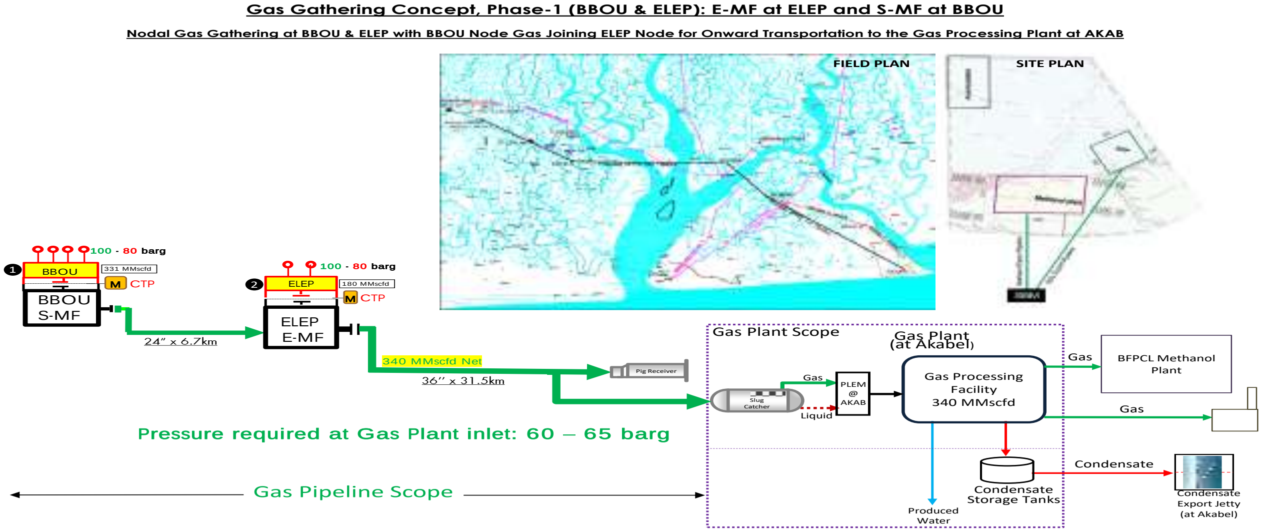

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this specification is to provide the minimum requirements that relate to the design, manufacture, and supply of electrical equipment used for the fabrication of electrical services to achieve connectivity for phase 1 of BFPCL Gas Gathering Pipelines & Associated Infrastructure.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
IEC 60086	Primary Batteries
IEC 60904	Photovoltaic Devices
IEC 61523	Delay and Power Calculation Standards
IEC 61537	Cable management – Cable tray systems and cable ladder systems
IEC 61727	Photovoltaic (PV) systems – Characteristics of the utility interface
IEC 61730	Photovoltaic modules
IEC 62108	Concentrator photovoltaic (CPV) modules and assemblies
IEC 62109	Safety of power converters for use in photovoltaic power system
IEC 62305	Protection Against Lightning: General principles
IEEE 1144	Recommended Practice for Sizing Nickel-Cadmium Batteries for Photovoltaic (PV) Systems
IEEE 1562	Guide for Array and Battery Sizing in Stand-Alone Photovoltaic Systems
BS EN 50521	Connector for photovoltaic system. Safety requirements and tests
BS EN 50618	Electrical cables for photovoltaic systems
IEC 60050	International Electrotechnical Vocabulary
IEC 60364	Electrical installations of buildings
IEC 60446	Wiring colours
IEC 60502	Power Cables with extruded insulation and their accessories for rated voltages from 1 kV (Um =1.2kV) up to 30 kV (Um =36kV)
IEC 60529	Degrees of protection provided by enclosures (IP Code)
IEC 61000	Electromagnetic compatibility (EMC)
IEC 61439	Low-Voltage distribution board and control gear assemblies
IEC 61641	Enclosed low-voltage distribution board and control gear assemblies
IEC 60898	Electrical Accessories. Circuit Breakers for Overcurrent Protection
IEC 61643	Surge protective devices connected to low-voltage power distribution systems
IEC 60598	Luminaires
ISO 9000	Quality Management System
NFPA 70	National Electrical Code

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. General Information

3.1. General Description

For the purpose of FEED, power supply to the manifold stations shall consider two most viable options listed below,

3.2. Option 1

Use of underground power cable from Akabel Power Plant to AKABEL Manifold station and further extension by use of RMU to ELEPA & BBOU. This option involves laying of underground cable (single or Composite) from Akabel to BBOU a total distance of circa 38.5KM with the aid of extensible RMUs and step-down to 400/230VAC at the utilization stations. This option will guaranty steady power supply and will eliminate use of generators or any other form of supply prone to vandalism.

3.3. Option 2

Option 2 involves use of off grid solar inverter system with back-up generators designed with auto start and stop with manual override as emergency back-up power supply to the Manifolds. One major challenge is need for a dedicated non-classified area for the installation of solar panels

The photovoltaic power supply systems shall supply continuously output voltage of 24 VDC and 400/230 VAC (through an inverter) during operation.

Each photovoltaic power supply system shall consist of the following main parts:

- 100% rated photovoltaic module subarrays,
- 100% rated charge controller,
- 100% rated inverter,
- 100% rated maintenance free Lithium-ion Phosphate (LiFePO4)) batteries,
- 24 VDC distribution board,
- 400/230VAC distribution board,
- junction boxes,
- circuit breakers,
- cabling,
- supporting structures,
- protection against vandalism such as solar panel clamps or any other anti-theft device recommended by vendor.

All photovoltaic system components shall be of the Vendor's best quality and construction to ensure full operability during project station life.

The high operating temperature, strength and frequency of storms and sand effects must be considered in the design.

Arrays, batteries and controllers shall be designed and arranged in such a way to provide easy extension for any future requirements. System shall be capable for continuous unmanned operation. System design shall include lightning protection and PV array shall have anti-bird protection for roof mounted PV array. In addition, surge arrestors shall be installed for devices sensitive to overvoltage's.

3.4. Photovoltaic Equipment Sizing

The photovoltaic power supply system shall be sized based on Load Lists provided for this station, single line diagram and other specific requirements covered within this specification.

All components sizing shall include 1.1 safety factor. The sizing of Photovoltaic modules shall be based on the consideration of the following factors:

- location insolation,
- number of modules necessary for proper output,
- internal power consumption,
- modules efficiency,
- battery daily and full recharge current,
- load current consumption,
- derating factors:
  - meteorological uncertainties,
  - ambient temperature,
  - offset from ideal horizontal and vertical angle,
  - lifetime degradation,
  - shading.

The sizing of Storage battery shall be based on the consideration of the following factors:

- battery autonomy for 4 days (96 hours), with 100% dual redundancy but operating in 50% load sharing mode.
- maximum depth of discharge at 80% level,
- ageing factor of 0.8,
- load voltage tolerance,
- type of float and high-rate chargers selected.

Each system shall be provided with automatic battery charging regulator / controller with measurement, monitoring and protection utilities.

4. Low Voltage Distribution Board

4.1. Technical Data

4.1.1. Main Bus Bar

The phase, neutral and earth busbars shall be of hard drawn high conductivity copper. The bus bars shall be selected according to the electrical requirements of IEC 61439 and IEC 60947. Bus bar joints shall be firmly secured with cover that is at least IP 42 and has captive bolts and nuts. The busbar system shall be accessible for construction and maintenance duties. The neutral bar shall be insulated from earth and extended to all compartments reached by the phase busbars. The earth bar shall be located in the bottom compartments over the whole length of the DB and in all cable, riser compartments (top to bottom) and easily accessible. Sufficient connection points with adequate terminating facilities shall be provided for terminating the cable earth leads and the external earth connections. For the termination of external earth connections, the earth bar shall be provided with an M10 bolt near the bottom of each outgoing cable compartment and at the incoming panels.

4.2. General Requirements

4.2.1. Basic Requirements

Distribution boards covered by this specification includes AC / DC Distribution Board for distribution of power supply as well as lighting and small power Distribution Board. LV distribution Board shall be designed for a lifetime of 30 years. LV distribution board shall be as follows

- Self-standing (metal enclosed assembled with single bus bar system and fixed mounted facilities)
- Wall Surfaced mounted
- Recessed / Flushed Mounted
- Stanchion Mounted

DB shall be type-tested assemblies as defined in IEC 61439-1. Copies of certificates and details of type test carried out shall be provided with the offer.

All materials and parts comprising the equipment shall be new and unused, of current manufacture, of the highest grade and free from all defects and imperfections affecting performance.

Low voltage distribution board shall be designed and manufactured with the highest quality standard of workmanship to ensure:

- Personnel health and safety.
- Safety for equipment and plant.
- Minimal fire risk.

- Reliability and continuity of service.
- Simplicity of operation.
- Flexibility of operation.
- Maintenance (accessibility and availability for inspection and repairing).

Any deviation from these requirements and relevant attachments shall be clearly listed during the bidding stage. In absence of a list of deviations, it will be assumed that the MANUFACTURER complies with all documentation.

4.2.2. Location

The location of the LV distribution board covered under this specification shall be indoors inside air-conditioned rooms and outdoors under sheds. The DB shall be installed both walls mounted and self-supporting as per requirement.

4.2.3. Tropicalization

LV distribution board assembly shall be tropicalized in order to eliminate mildew, fungi and other detrimental effects of tropical environment.

4.2.4. Main Power Supply

The nominal supply voltage and frequency to the Low Voltage distribution board shall be 400/230 V AC +/- 10% and 50 Hz +/- 2%. For DC distribution board, power supply shall be 24 V DC.

4.2.5. Cable Entry and Gland Plate

Cable entry shall be from bottom for main LV distribution board. For all other sub DB/dedicated DB of each building, cable entry can either be from top or bottom (top entry shall be preferred). For outdoor DB installed in shed, bottom entry shall be preferred. Gland plate shall be made of non-magnetic metallic plate with sufficient thickness to provide mechanical strength (minimum thickness of 4mm). The gland plate shall be earthed to the LV distribution frame.

4.3. Internal Wiring

The LV distribution board shall be wired as per IEC 60445 standards, especially providing protection against contact with live parts. All power wiring shall be rated at 1000 V. Cables shall be rated for 90 °C conductor temperature with flame retardant insulation. Cable core shall be stranded copper, class 2, according to IEC 60502. Cables between two connecting devices shall have no intermediate splices or soldered joints. Connections shall, as far as possible, be made at fixed terminals according to IEC 61439. The wires shall be selected according to load, insulation level, function and operating requirements. The wires shall be adequately rated and equipped with fuses / circuit breakers of adequate size.

Provision to protect the wiring against any mechanical damage shall be considered. Wires shall be bundled and fixed.

All wires shall be terminated to the internal side of terminal blocks. Not more than two wires shall be terminated in the same terminal, in such case double compression type ferrule will be used to join two conductors in a common end. The wiring between two internal terminals shall be continuous, no joints are allowed. The identification of wires shall be unambiguous through labelling at both ends on which the origin and the destination of terminal blocks and of terminal strips shall be marked in indelible way.

4.4. Mechanical Requirements

4.4.1. Construction Requirement for Distribution Board

The DBs shall be flush / surface / stanchion mounted. It shall be TPN DB with single or double door depending on size.

The board shall comprise:

- A distribution panel for housing the MCBs.
- Copper Busbar, short links, neutral and earth bar
- Installation accessories like DIN rails for mounting MCBs,
- Connection accessories like common busbars to simplify interconnection of devices, accessories for fixing conductors, terminal blocks for cable termination, neutral busbar and earthing bar, etc.
- Service cable ducts or conduits, surface mounted or in cable chases embedded in the wall.

The board should be installed at a height such that the operating handles, indicating dials (of meters) etc., are accessible to users. Recommended height for installation in buildings is 2 meters.

4.4.2. Degree of Protection for the Assembly

- Ingress Protection

The LV distribution board assembly shall be completely enclosed with IP 42 degree of protection as show in table 6.0 below:

For electrical equipment in outdoor areas.	IP 55 minimum
For distribution boards, junction boxes, UPS enclosure, switchgear and panel boards in indoor areas	IP 42 minimum
For indoor fittings like sockets, switches, etc	IP 23 minimum
For indoor fittings installed in wet locations (toilet, bathroom & kitchen)	IP 55 minimum

Table 6.0: Electrical equipment minimum degree of ingress Protection

- Protection against Electrical Shock

The LV distribution board shall be designed to ensure protection against inadvertent electric shock. This shall comprise measures for both direct and indirect contact. Protection against direct contact shall be ensured by insulation of live parts and the use of barriers. All exposed surfaces shall conform to a degree of protection against direct contact of at least IP23. All barriers shall be firmly secured in place. The live parts inside the fixed compartment shall be protected against touch with a degree of protection of at least IP 23. Barriers shall be put in place to prevent danger to personnel working on a disconnected functional unit with adjacent units still in operation.

4.4.3. Distribution board Enclosure

The sheets shall have a minimum thickness of 2 mm with reinforcements at weak points, except for internal sheets not belonging to the bearing structure and of the external enclosure, which may have a thickness of 1.5 mm. The closing covers and plates shall be fixed by captive bolts, nuts or screws.

Enclosure of the distribution board shall have a resistance to mechanical shocks. The type of keys and tools shall be reduced to a minimum consistently with the safety requirements.

4.5. Electrical Requirement

4.5.1. Control Voltage

The control voltage where applicable shall be 230 V AC.

4.5.2. Control Switches

The LV distribution board feeding external lighting /outdoor circuit lighting shall be equipped with "Auto/Manual" selector switch which shall allow for photocell auto operation for selected outdoor lighting circuits. In manual position the photocell operation shall be bypassed. The selector switches shall withstand operational switching for the lifetime of the LV distribution board.

4.5.3. Moulded Case Circuit Breaker

MCCB shall be TPN+E type. Incomer MCCB shall be hand-operated and shall be thermal-magnetic type.

4.5.4. Miniature Circuit Breaker

MCB shall be SPN type. Miniature Circuit Breaker (MCB) can be used up to a size of 63 A. For rating above 63 A, MCCB shall be installed.

4.5.5. Residual Current Operated Circuit Breaker (RCBO)

RCBOs with 30mA RCD protection shall be used to provide additional overload protection and shall be applied to socket outlets, AC circuits, outdoor lighting circuits.

4.5.6. Surge Protective Devices (SPD)

Surge devices shall be provided and installed in accordance with IEC 60364-4-44 "Low voltage electrical installations – Part 4-44: Protection for safety – Protection against voltage

disturbances and electromagnetic disturbances." Surge devices shall be installed near the origin of the LV distribution service supply.

5. Cables

5.1. Electrical Cable Requirements

Cables shall be capable of continuous operation at the highest system voltage as specified above with a maximum conductor operating temperature of 90ºC and a maximum temperature under fault conditions of 250°C. Current rating of cables shall be in accordance with IEC 60287. Each layer of material shall have a smooth external surface free of indentations, ripples etc. Cables shall be designed and constructed to provide a design life of 25 years in the environment and operating conditions specified.

5.2. Construction Requirements

5.2.1. Conductor

The conductor for LV cable shall be of high conductivity plain annealed stranded copper wire according to IEC 60228. The class of conductor shall be Class 2 for power cables and control cables.

5.2.2. Insulation

The conductor insulation shall be Cross Linked Polyethylene (XLPE) compounds (IEC 60502-1 & 60502-2). The insulation for LV cables shall be rated for 0.6/1.0 kV. For earthing conductors and for protective conductors the insulation shall be PVC as per IEC 60502. The insulation for earth cable shall be rated for 450/750V. The insulation colour for the earthing conductor PVC shall be yellow/green.

5.2.3. Filler

Fillers shall be applicable for multi core cable. Fillers shall be non-hygroscopic materials suitable for the operating temperature of the cable and compatible with the insulating material as per IEC 60502.

5.2.4. Inner Sheet

Inner sheath shall be extruded type and shall be made of Polyvinyl Chloride (PVC) as per IEC 60502. This is applicable for outdoor cables which shall have armour and outer sheath covering.

5.2.5. Bedding

Over the metallic sheath shall be applied a layer of bedding which shall serve to protect the metallic sheath against corrosion and from mechanical injury due to armouring. Suitable material like jute or hessian material may be used.

5.2.6. Armour

Armour shall consist of Steel wire Armour (SWA) for Multi core cable and Aluminium wire Armour

(AWA) for single core cable. The dimension of armour shall be in accordance with IEC 60502-1.

5.2.7. Outer Sheet

Outer sheath shall be of extruded type and shall be made of Polyvinyl Chloride (PVC) as per IEC 60502.

5.2.8. Special Components & Construction

The LV cables shall be constructed with the below requirement as a minimum

- Flame Retardant (IEC60332-1)
- Fire Retardant (IEC60332-3)
- Termite and toredo worms attack protection type
- Sunlight (UV) resistance

5.2.9. Core Identification

The core identification of the power and lighting cables shall be as shown in the following table in accordance with IEC 60445:

Single or multi-core cable	Application	Core identification colour
Single-core conductor (Phase L1 and L2 and L3)	Phases	Black
Three-core cables or circuit (AC. single phase + PE)	Phase	Brown
	Neutral	Blue
	PE	Yellow/green
Three-core cables (AC. three phase system)	Phase L1	Brown
	Phase L2	Black
	Phase L3	Gray
Five-core cables or circuit (AC. three phase + neutral +PE)	Phase L1	Brown
	Phase L2	Black
	Phase L3	Gray
	Neutral	Blue
	PE	Yellow/green
Four-core cables (AC. three phase + PE)	Phase L1	Brown
	Phase L2	Black
	Phase L3	Gray

Single or multi-core cable	Application	Core identification colour
	PE	Yellow / green
Two-core cables or circuits (DC. system)	Positive Negative	Brown Gray

Table 7: Cable and Conductor Core Identification

5.2.10. Design Requirements

All fillers, inner and outer sheaths also bedding under armour wires shall be of the reduced toxicity type. The material shall be non-hygroscopic and have flame retardant properties, with an oxygen index of 33 or a numerically higher value. Maximum levels of hydrogen chloride (HCl) emission for sheath material shall not exceed 8% by weight for black (and 9% by weight for other colours). Cables shall be tested according to IEC 60754. The cable insulation, bedding, sheath and overall sheath shall each be applied symmetrically. The bedding, inner sheath and outer sheath shall be circular. The maximum difference in diameters measured at different points around the circumference and along the cable shall be 0.2 mm. The maximum difference in radial thickness of material shall be 0.2 mm.

5.2.11. Bending Radius

All cables shall be capable of bending to a minimum radius of six (6) times overall cable diameter without affecting their electrical characteristics in any way or causing the armouring to “birdcage”.

5.2.12. Cable Joint

Cables shall be supplied in continuous drum lengths without any cable or conductor joints. Cable joint shall be avoided to the extent possible. However, for long cable lengths cable joint shall be acceptable subject to written approval by COMPANY

5.3. Cable Types

5.3.1. Power Cables for low voltage

Power cable shall either be single core or multi core cable based on the following criteria. For power cables up to185 mm^2 conductor cross section shall be multicore, while cross sections exceeding 185 mm^2 shall be single core. The minimum cross section of low voltage power cables shall be 2.5 mm^2.

- 2 Core LV Cable

These cables shall be used for power connection between DC power supplies.

- 3 Core LV (Ph+N+PE) single phase

The cable indicated on this paragraph shall be used for lighting system and all external single-phase Loads.

- 4 Core LV (3Ph + PE) three phase

The cable indicated on this paragraph shall be used for LV three phase motors.

- 5 Core LV (3Ph + N + PE) three phase

The cable indicated on this paragraph shall be used for main distribution board system (Three phase), Streets and Fence Lighting.

The construction for power cables shall be as follows

Cable Type	Indoor	Outdoor
Multi Core Power cable	CU/XPLE/PVC	CU/XPLE/PVC/SWA/PVC
Single Core Power cable	CU/XPLE/PVC	CU/XPLE/PVC/AWA/PVC
Grounding cable	CU/PVC	CU/PVC

Table 8: Cable Type Definition

6. Lighting Fixtures

All lighting fixtures to be used for both indoor and outdoor areas, hazardous & non-hazardous areas shall be of LED type with appropriate IP & ATEX classification.

6.1. Technical Requirements for Lighting Fittings

All lighting circuits shall include Miniature Circuit Breakers (MCB), switches and junction boxes as required.

The internal wiring cable for the lighting fixtures shall be flame retardant: IEC 60332 Part 3. The Vendor shall state to which standard the cables offered are manufactured.

Lighting fixtures shall be suitable for both external and internal installations and pole mounted, heavy duty with weatherproof head. Internal connections shall be made by anti-loosening devices or terminals. The degree of protection of the fixture and cable glands shall be IP65. Every single component of fixture shall be provided with a mark indicating the Vendor's identification, protection degree & safety execution. The internal reflector shall be protected by thermal shock resistant glass. The materials shall be tropicalized with readily available spares in Nigerian market.

All equipment directly exposed to atmosphere shall be weatherproof, suitable for tropical savannah (wet and dry) environment, and have a minimum degree of ingress protection IP65, in line with IEC 60529. All outdoor equipment and junction boxes in hydrocarbon environment must be Atex compliant.

All equipment for surface mounting shall have robust external fixing lugs individually or in combination.

The offered equipment shall be suitable for operation in exposed outdoor locations under the conditions as specified and the materials employed in construction selected to prevent corrosion.

The colour and finish shall be in accordance with the Vendor's standards for the service conditions specified.

The equipment shall be afforded adequate protection against mechanical damage, deterioration, etc., until it is unpacked at site.

All threaded entries shall be ISO Metric type in accordance with BS 3643, R68 series constant pitch 1.5 mm, Class 6g fit. All entries shall be plugged with certified plugs to suit equipment classification and to prevent the ingress of dirt and moisture. Some of the plugs will remain when equipment is in operation.

Danger and warning labels shall be written in English language. Identification labels shall be in English.

For indoor lighting, packages shall be suitable for non-hazardous area. Emergency lighting if required will be in accordance to BS 5266-1. The photocell for daylight monitoring shall be industrial (230V/50 Hz) suitable for use in hazardous environment with ingress protection of IP65. It shall be made of plastic material.

All outdoor lighting fittings shall be suitable for installation within a zone 1 and/or Zone 2 hazardous area (ATEX category 2) with certification ranging from TEEX "e", EEx ''d'' or EEx ''de'' Group IIC, Temperature Class T3 depending on the classification of the location and IP65 Junction boxes installed adjacent to the lighting poles.

Certified lighting equipment shall not be tampered with, modified or drilled in any way whatsoever except with consent from the COMPANY.

Material specifications for poles and accessories shall be of the following unless overridden by Structural specification:

- Steel Work - S275JR BS EN 100025:2004
- Bolts - Grade 8.8 BS 3692:2001
- Nuts - BS3692:2001

6.2. Illumination Levels

Illumination levels of indoor and outdoor lighting shall be as specified in table 8 below:

Location	Maintained Illuminance E_m (Lux)		Unifor-mity Factor (U)	Glare Rating (UGR_L/^RGL)	Colour Render-ing Index (Ra)	Remarks
	Task Area	Surround-ing Area				
General Process Area	50	NA	0.4	45	20	
Compressor, Generator and MV Motor, Pumps Exchangers, Separator Areas	100	50	0.5	45	40	
Stairways & Ladders	100	50	0.25	45	20	
Local Control Panel	200	50	0.5	45	40	

Road, Fence, Walking Areas Lighting	10	10	0.25	50	20	
Car Park	20	20	0.25	50	20	
Junction Points / Security Posts	50	20	0.4	45	20	
Other Notified Outdoor Areas	10	10	0.4	45	20	
Muster Areas	100	50	0.4	45	20	
Electrical Rooms	300	300	Note 1	22	80	
Central Control Room	500	300	Note 1	16	80	Light should be controllable
Battery Rooms	200	200	Note 1	22	80	
Offices	500	300	Note 1	19	80	
Bedroom	100	100	Note 1	22	80	
Bathrooms	200	200	Note 1	25	80	
Stairs	150	150	Note 1	25	40	
Corridors	100	100	Note 1	28	40	

Table 9: General Requirement for Lighting

Note 1: See table 10 below for uniformity factor for indoor areas.

Task Illuminance Lux	Illuminance of Immediate surrounding areas Lux
≥750	500
500	300
300	200
≤200	E_{task}
Uniformity: ≥0.7	Uniformity: ≥0.5

Table 10: Indoor Lighting Uniformity Factor

7. Earthing

7.1. General Requirement

All Earthing and Lightning Protection System materials shall be brand new and manufactured of the highest grade and free from all defects and imperfections affecting performance. Vendor shall be responsible for design, engineering and manufacturing of the equipment to fully meet the intent and requirements of this specification.

All accessories required for completeness of every Earthing and Lightning Protection System, whether specifically mentioned or not, but considered essential for satisfactory performance shall be included as a part of the offered item. All Earthing and Lightning Protection System shall be designed for a lifetime of 30 years.

All Earthing and Lightning Protection System shall be designed and manufactured with the highest quality standard of workmanship to ensure:

- Personnel health and safety.
- Safety for equipment and plant.
- Reliability and continuity of service.
- Simplicity of operation.
- Flexibility of operation.
- Maintenance (accessibility and availability for inspection and repairing).

Any deviation from these requirements and relevant attachments shall be clearly listed during the bidding stage. In absence of a list of deviations, it will be assumed that the VENDOR complies with all documentation.

7.1.1. Location

The location of the Earthing and Lightning Protection System covered under this specification shall be both indoors and outdoors.

7.1.2. Tropicalization

The Earthing materials shall be tropicalized in order to eliminate mildew, fungi and other detrimental effects of tropical environment.

7.1.3. Contingency

All Earthing and Lightning Protection System shall be considered for 20 percent spare to cater for contingency during construction and commissioning.

7.1.4. Mechanical Strength

All Earthing and Lightning Protection System and accessories shall be of a mechanical strength sufficient to withstand the stresses to which they may be subjected in normal service.

7.2. Technical Requirement

TN-S earthing system shall be provided for the stations. The earthing network shall consist of main grid and branch connection. The earthing network shall be made of bare copper conductor with minimum cross section 70mm² with a sufficient number of distributed earthing rods in the station. Size of bare copper conductor and number of earthing pit shall be confirmed by carrying out Earthing calculation for the station.

Earth grid shall consist of insulated earth conductors, vertical earth electrodes, and shall be installed homogeneously distributed over the network grid with a spacing depending on soil resistivity.

Earthing conductor shall be laid directly in the ground. Underground joints shall be compression or cadweld type. Main earthing bars shall be provided above ground to allow attaching of bonded equipment. Earthing bars shall be mainly installed on metallic structures. Earthing bars shall be distributed throughout the stations to minimize bonding connection lengths.

All connections of conductors on equipment shall be performed with pressure type lugs or connectors and threaded bolts, screws, spring-washers and washers. Special care is to be taken to avoid the arising of a chemical element.

Connections between bare copper and iron parts are to be protected in a special manner and shall only be executed on above-ground connection points or inside of pits.

7.2.1. Construction Requirement for Main Earthing System

- Main Earthing Grid Conductor

The main earthing conductor shall be of high conductivity plain annealed stranded bare copper wire. The class of conductor shall be Class 2. Conductor shall be tested in accordance to IEC 60228. The conductor shall be designed for service life of 30 years. The size and continuous current carrying capacity of the conductor shall be based on a maximum permissible temperature considering the solar radiation and wind effects. The bare copper conductors shall consist of concentric-lay stranded conductors made from uncoated annealed round soft drawn copper wires of class 2 non-compacted. No joints of any kind shall be made in the finished copper wires. The main earthing grid conductor shall be laid at least 50 cm below final grade in natural soil. In case of common laying of earthing network and cables in one cable trench the conductor shall be placed beside the cable route and surrounded with compacted material prior to sand bedding necessary for the cable laying.

- Green/yellow Insulated Earth Cable

For insulated earthing cables, the insulation shall be made of PVC as per IEC 60502 and shall be rated for 450/750V. The insulation colour for the earthing conductor PVC shall be yellow/green. The conductor shall be of high conductivity plain annealed stranded copper wire. The class of conductor shall be Class 2. Earth cable shall be flame retardant in accordance with IEC 60332-1-2 and IEC 60332-3. Conductor shall be tested in accordance IEC 60228. Earth cable shall be hydrocarbon resistance, anti-termite, anti-teredo worm and Ultraviolet resistance. The size and continuous current carrying capacity of the conductor shall be based on a maximum permissible continuous temperature of 70°C. Equipment bonding shall be by the use of green/yellow PVC insulated earth cables of different sizes. Green/yellow PVC insulated earth cables of 70mmsq minimum cross section shall also be used for the connection between inner and outer earthing rings of buildings. Refer to Table 8 for PVC insulated earth cable sizes for different applications.

- Exothermic Welding Materials

Connection of Main and branch buried grounding conductors shall be realized through exothermic welding to achieve quality electrical earth bonding. The weld powder material mixture for the exothermic weld shall be copper oxide and aluminium. Thermal welding shall be moulded fusion welding process by cad weld. The accessories shall include:

7.2.2. Copper Earth Tape

Copper earth tape shall be used as lightning down conductor for all building in the station. High conductivity, annealed, soft drawn flat copper earthing tape, manufactured and tested in accordance with IEC 62561 suitable for use within a variety of earthing (grounding) and lightning protection applications. Copper earth tape shall be 25 mm width x 3 mm thick.

7.2.3. Earthing Rod

Earthing rods shall be copper (Cu), Ø 15 mm, with minimum length of 3 m of the extensible type copper coupling material complete with galvanized steel driving tip and cover. The driving tip and the head cover shall fit the Ø 15 mm earth rod.

7.2.4. Earth Pit

Rectangular chamber made of concrete shall be used as Earth pit. Earthing pit shall be complete with concrete cover to provide protection and access to the clamped connection of the ground rod. Earthing pit concrete chamber dimension shall be length = 450mm, width = 450mm and depth = 400mm. The concrete cover shall be length = 350mm, width = 350mm. The cover shall have a lifting handle. The concrete housing shall have bottom opening for incoming and outgoing grounding cables. Earthing pit shall be tagged according to the project tagging philosophy. The cover shall be engraved with an earthing symbol to indicate its functionality. The earth pit shall house the earth rod. Earthing rods shall be copper (Cu), Ø 15 mm, with minimum length of 3 m of the extensible type copper coupling material complete with galvanized steel driving tip and cover. The driving tip and the head cover shall fit the Ø 15 mm earth rod.

The individual earth electrode of respective earth pit shall connect to the earth grid using connecting earth bar which serves as an isolating feature for disconnection during testing. The earth electrodes shall be tested periodically according to program scheduled by COMPANY to verify the earth resistance between the electrode and true earth. And to achieve this test, the earth grid shall be isolated from the grounding electrode under measurement by disconnecting it from the rest of the system via the connecting earth bar. The test shall be conducted using insulation tester megger equipment.

7.2.5. Earth Boss

Earth boss shall be made of steel. Earth boss shall be used for earth connection on steel structures. The dimension of the earth boss shall have diameter Ø 40mm with a thread of M10 X 30mm.

7.2.6. Earth Bar

Earth bar shall be hard copper drawn and shall be used both indoor and outdoor. Hole size for termination shall be M10. Thickness of the earth bar shall be 6mm. The accessories shall be complete with M10 hexagon head screws, nuts and washers. The below dimension shall be used as applicable and as required.

8. Marking

All labels and marking shall be in English language. All components, equipment and installations shall be clearly identified with permanent descriptive labels that facilitate easy recognition by the operator which shall be of stainless-steel type 316L or equivalent non-corrosive material. Tagging plates or labels on fronts of enclosures should be fixed with screws. Adhesive nameplates shall not be used.

- A rating nameplate for electrical equipment
- COMPANY's Name,
- Manufacturer's Name and Address,
- Purchase Order Number, Serial Number and Year of Construction,
- Equipment identification number,
- Rated voltage, current, frequency, short circuit current,
- Manufacturer's shop reference and drawing number,
- IEC certification.

Equipment Nameplates shall be black lettering on white background. Warning nametags shall be white lettering on red background.

All control components shall be identified with permanently installed nameplates showing equipment control and devices function. Terminals shall be sequentially numbered.

9. Quality Assurance

Equipment manufacturer shall identify the quality control standard adhered to at his facility. Where no specific quality control standard is followed, manufacturer shall provide sufficient details of his quality control program to determine compatibility with other recognized standards. Manufacturer shall provide access to his premises for quality surveillance by the COMPANY / Purchaser as necessary.

10. Painting

All metallic structures shall be adequately protected against corrosion. Panels shall be sand blasted, primed and painted with two coats of grey epoxy or enamel as per the manufacturer's standard. Touch paint shall be supplied with assembly. Damages, caused by transport, installation, cabling or commissioning shall be repaired in a way, that the original quality is restored. Additional painting at location of installation shall not be allowed.

11. Inspection and Testing

11.1. Factory Acceptance Tests

Full functional tests requirements as defined in IEC 61724, IEC 62040-3 shall be carried out. A formal Factory Acceptance Tests of photovoltaic power system elements and LV Distribution Boards shall be carried out. The Supplier shall specifically produce a detailed Inspection Test Plan. All functions shall be tested. Where communications are required to third party equipment, tests may be carried out separately to prove the communications and functions operate correctly. Each test shall be witnessed by COMPANY representative.

11.2. Site Acceptance Tests

A formal Site Acceptance Test shall also be performed with COMPANY representative and commissioning engineer support. Integration tests shall verify all system interfaces. A reliability test shall also be performed at site following a successful SAT. The format and duration of this test shall be discussed and agreed with COMPANY.

The Supplier shall carry out full functional tests on the system using a dummy load to represent the actual average daily load anticipated.

Particular attention shall be paid to monitoring the output of the photovoltaic panels including current balance, and output variation according to time of day and for different angles of inclination and at different temperatures.

A satisfactory degree of conformity with design figures shall be demonstrated.

11.3. Type Tests

Type tests shall not be performed, if the vendor submits a copy of the type test certificates for similar type and rating. If not available, type test shall be carried out on one identical unit

12. Documentation

Complete documentation shall be provided for the design, manufacturing, testing, commissioning, start-up, operation, maintenance and repair of system components.

Documentation has to be prepared in accordance with the relevant ISO standards or in the absence of relevant details in those standards the DIN standards shall apply.

The final documentation shall be delivered on paper in sufficient number and in electronic form.

13. Spare Parts And Special Tools

Supplier shall deliver all spare parts, which are necessary for commissioning, start-up and one-year operation time.

All necessary special tools for operation and maintenance of each panel shall be supplied and at least consisting of:

- One set of all special tools required for operation and maintenance of the electrical system equipment,
- One set of circuit breakers (2 pieces of each type),
- Key set for cabinet doors and locks,
- Magnetic warning labels in sufficient number

14. Shipping

All equipment, material and spare parts shall be sufficient packaged. Each component should be shipped as a common delivery. The material, especially loose parts and spare parts shall be clearly labelled according to the designation.

Items easily damaged during shipping shall be removed and packaged separately. All equipment shall be packaged for outdoor storage. The Supplier shall recommend handling requirements and ship the equipment accordingly. At least one copy of assembly drawings in a weatherproof packet shall accompany the equipment when shipped.